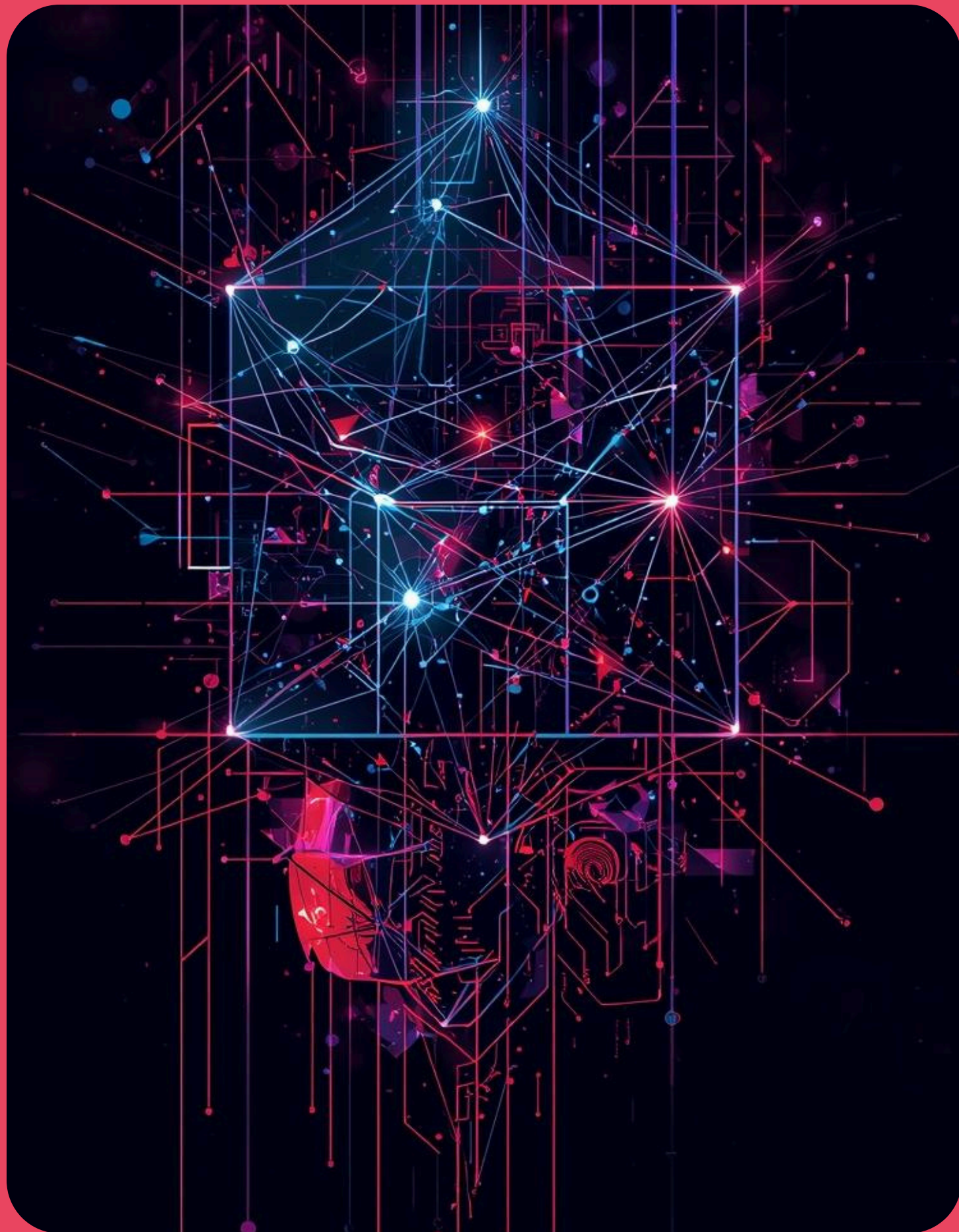
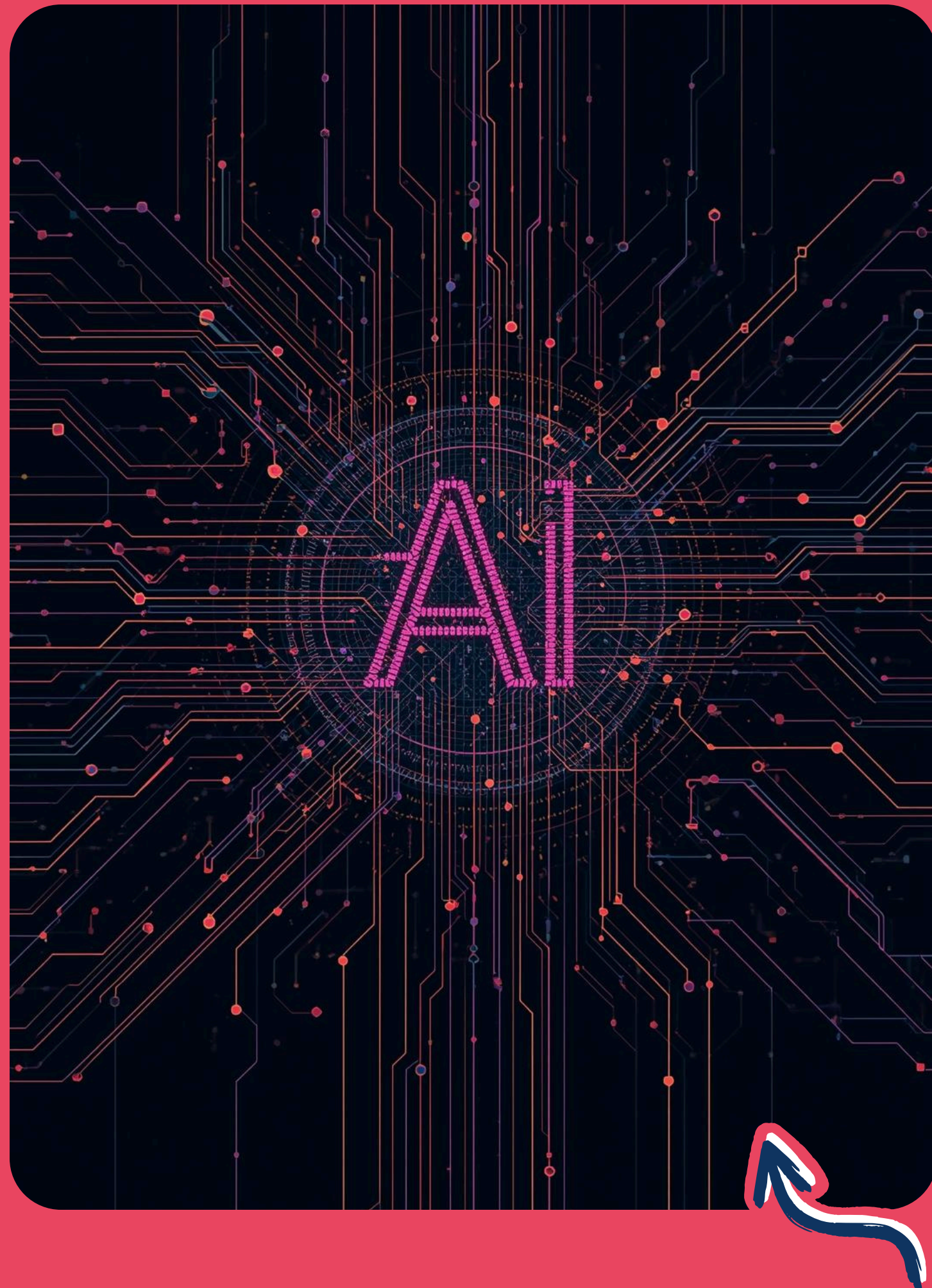




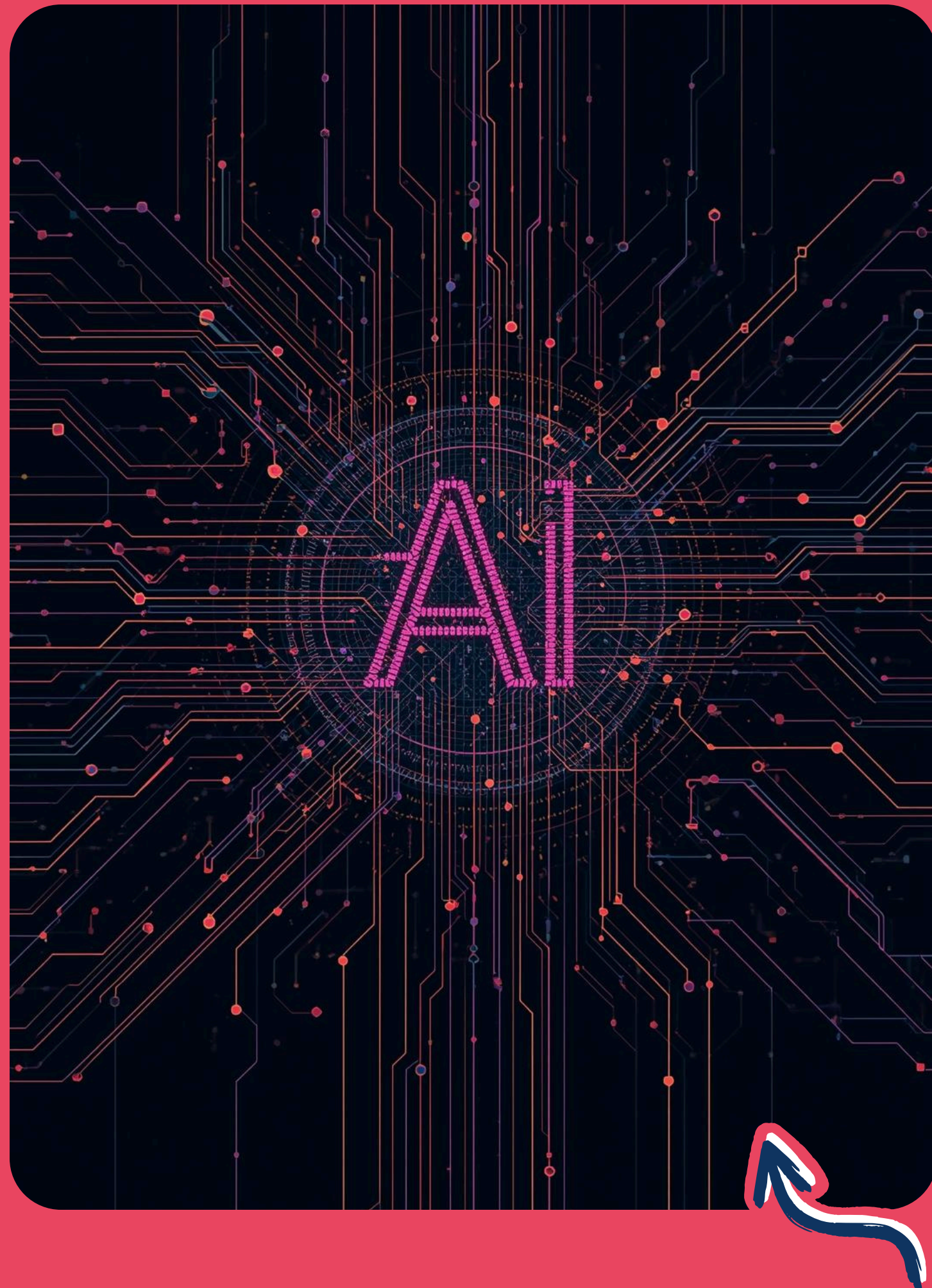
AI as a Black Box Problem



The Black Box Problem

The black-box problem refers to AI systems that produce outputs or decisions without a clear, understandable explanation of how those decisions were made.





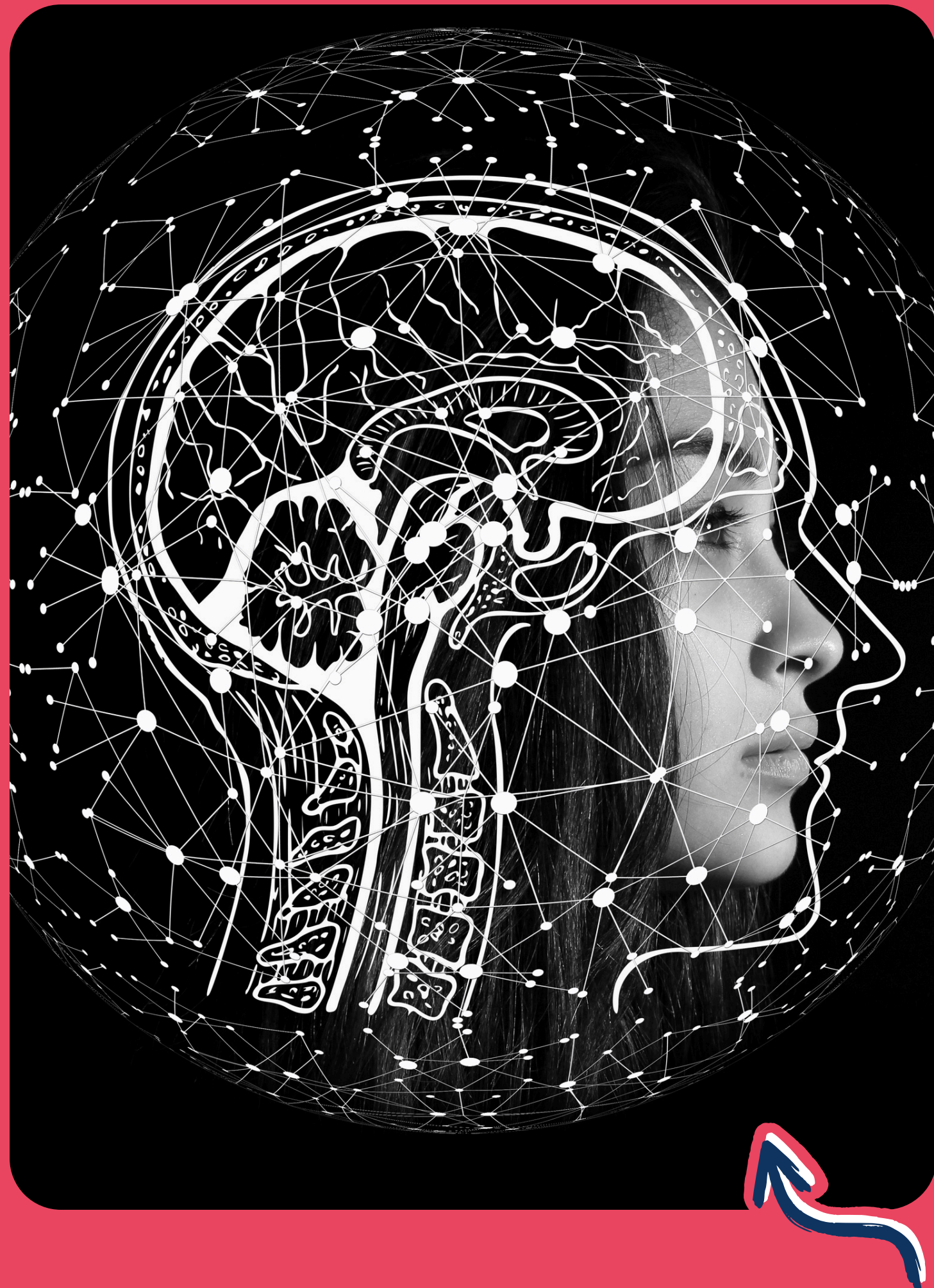
Why this Happens

Black-box behavior is common in:

- Deep learning systems
- Large neural networks
- Highly complex models trained on massive datasets

These systems often prioritize accuracy over interpretability.

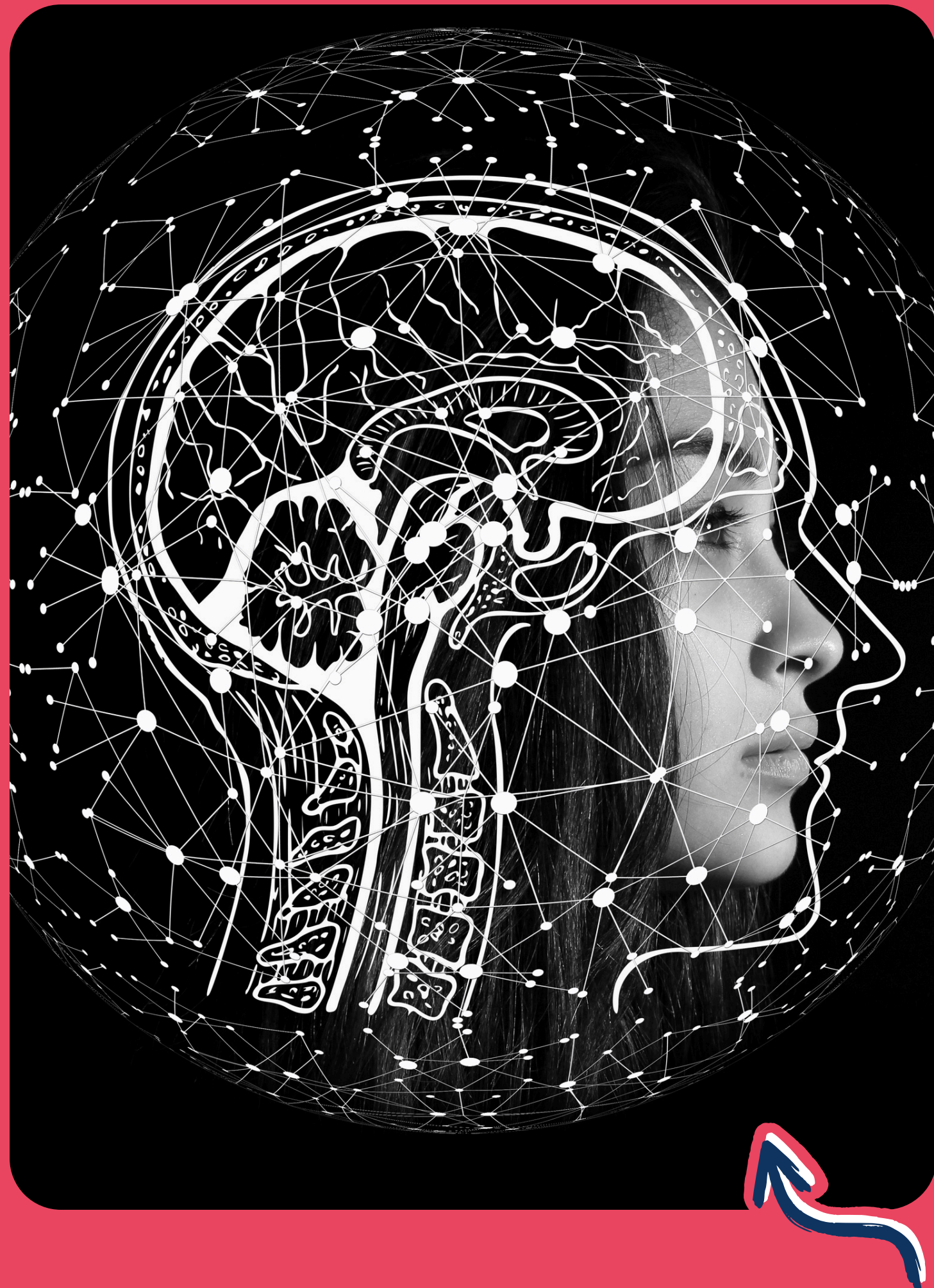




An Example

- GPT-4: 1.76 trillion parameters (estimated)
- That's 1,760,000,000,000 numbers
- Impossible to trace how any single output was produced
- Emerges from interactions across billions of parameters
- Fundamentally beyond human comprehension





“Just look inside”

- Code is readable - it's basic neural network architecture
- But the learned weights? Just massive tables of numbers
- Like saying "just read all 1.76 trillion numbers"
- Numbers don't tell you what they represent
- Looking inside doesn't reveal meaning





Opacity

- Intentional
- Illiterate
- Intrinsic





Intentional

- Companies keep algorithms secret for competitive advantage (trade secrets)
- "Proprietary methods" = you can't inspect
- Even if you could access it, they won't share
- Legal protection for intellectual property
- Profit motive vs. public accountability





Illiterate

- Even with access, most people can't read code (expertise gap)
- Requires advanced math, computer science, statistics
- Creates dependency on experts
- Democratic deficit: can't challenge what you can't understand
- Technical gatekeeping





Intrinsic

- Even experts genuinely don't know how system works
- Not hiding anything - it's actually unknowable
- Emergent properties from billions of interactions
- This is the hardest problem
- May be impossible to fully solve

